

SVM Experiment Report: Effect of Penalty Parameter C on RBF-Kernel SVM

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1 Step 1: Dataset and Classification Task

This experiment uses the Iris dataset and keeps only two classes: *versicolor* and *virginica*. The selected features are petal length and petal width. The task is binary classification: predict whether an iris sample belongs to *versicolor* (label -1) or *virginica* (label $+1$).

versicolor and *virginica* were chosen because they are not linearly separable, which makes them more suitable for demonstrating the effect of a soft-margin RBF-kernel SVM than the easily separable *setosa* class. Two features are used so that the decision boundary can be drawn in two dimensions, satisfying the graphic-demonstration requirement.

A stratified 70/30 train-test split is used with `random_state = 42`. `StandardScaler` is fitted on the training set only and then applied to both training and test data. This prevents data leakage and ensures that the preprocessing is part of the model pipeline rather than a source of information from the test set.

2 Step 2: Methodology

2.1 Soft-Margin SVM with RBF Kernel

The model is a soft-margin support vector machine with an RBF kernel, implemented as `SVC(kernel="rbf", gamma="scale", C=C)`. The main experiment compares only the penalty parameter $C \in \{1, 50, 100, 1000\}$; all other settings are fixed.

To satisfy the requirement of using several methods, the report also includes a supplementary comparison of three SVM variants on the same train-test split: linear SVM, polynomial-kernel SVM, and RBF-kernel SVM. Linear SVM uses a straight separating hyperplane, polynomial SVM uses polynomial feature interactions, and RBF-SVM uses a nonlinear radial-basis similarity function. The RBF model is then used for the main parameter experiment because it gives the strongest test accuracy and F1-score among the three compared SVM methods in this setting.

The hard-margin SVM finds a hyperplane that maximises the geometric margin:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1, \quad i = 1, \dots, n. \quad (1)$$

When classes are not linearly separable, the soft-margin formulation introduces slack variables $\xi_i \geq 0$:

$$\min_{w,b,\xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad \text{subject to} \quad y_i(w^\top x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0. \quad (2)$$

The parameter $C > 0$ controls the trade-off between maximising the margin and minimising margin violations. A smaller C tolerates more violations and tends to produce a smoother boundary; a larger C penalises violations more strongly and fits the training data more

strictly.

The decision function for a kernel SVM is

$$f(x) = \text{sign}\left(\sum_{i=1}^n \alpha_i y_i K(x_i, x) + b\right), \quad (3)$$

where the RBF kernel is

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2). \quad (4)$$

With `gamma="scale"`, scikit-learn sets $\gamma = 1/(n_{\text{features}} \cdot \text{Var}(X_{\text{train}}))$, which gives $\gamma = 0.500000$ for this scaled training set.

2.2 Fixed Control Variables

The following settings are held constant across all four runs so that differences in results can be attributed solely to C :

- Dataset: Iris (`iris/iris.data`)
- Classes: *versicolor* (-1) and *virginica* ($+1$)
- Features: petal length and petal width
- Train/test split: 70%/30%, stratified, `random_state = 42`
- Preprocessing: `StandardScaler` fitted on training set only
- Kernel: RBF, `gamma = "scale"` (effective $\gamma = 0.5$)

2.3 Pseudocode

Input:

```
Iris dataset
Selected classes: versicolor, virginica
Selected features: petal length, petal width
C_list = [1, 50, 100, 1000]
kernel = RBF, gamma = scale, random_state = 42
train_test_split = 70/30 stratified
```

Procedure:

1. Load Iris dataset
2. Keep only versicolor and virginica samples
3. Keep only petal length and petal width
4. Encode labels as -1 and $+1$
5. Split data into training and testing sets (stratified)
6. Fit `StandardScaler` on training data only

7. Transform both sets with the fitted scaler
8. For each C in $[1, 50, 100, 1000]$:
 - a. Train `SVC(kernel='rbf', C=C, gamma='scale')`
 - b. Predict labels on training and test sets
 - c. Compute train accuracy, test accuracy, precision, recall, and F1-score
 - d. Compute the test-set confusion matrix
 - e. Record the number of support vectors
 - f. Plot decision boundary and support vectors
9. Summarise metrics in tables
10. Compare linear, polynomial, and RBF SVM on the same split
11. Compare figures and discuss the effect of C

Output:

Metrics table, support-vector table, confusion matrices, method-comparison table, decision-boundary figure (Figure 1), metrics-comparison figure (Figure 2), support-vector figure (Figure 3), discussion

3 Step 3: Experimental Results

3.1 Metrics Table

Table 1 reports the main performance metrics for each value of C . All numbers are obtained by running `svm_experiment.py` on the fixed data split described above.

Table 1: Main comparison of RBF-SVM models under different C values.

C	Train Acc.	Test Acc.	Precision	Recall	F1	# SV
1	0.9571	0.9000	0.9286	0.8667	0.8966	15
50	0.9857	0.9000	0.9286	0.8667	0.8966	9
100	0.9857	0.8667	0.8667	0.8667	0.8667	8
1000	1.0000	0.8667	0.8667	0.8667	0.8667	7

3.2 Support Vectors by Class

Table 2 breaks down the support-vector count by class.

Table 2: Support-vector counts for each class.

C	SV in <i>versicolor</i>	SV in <i>virginica</i>	Total SV
1	7	8	15
50	5	4	9
100	5	3	8
1000	4	3	7

3.3 Test-set Confusion Matrices

Table 3 reports the confusion matrices for the four main C values on the fixed test set. Each test set contains 15 *versicolor* samples and 15 *virginica* samples.

Table 3: Test-set confusion matrices for different C values.

C	ver. $\rightarrow$ ver.	ver. $\rightarrow$ vir.	vir. $\rightarrow$ ver.	vir. $\rightarrow$ vir.
1	14	1	2	13
50	14	1	2	13
100	13	2	2	13
1000	13	2	2	13

3.4 Comparison of Several SVM Methods

Table 4 compares three SVM variants on the same data split. This comparison is supplementary; the main parameter experiment still focuses on the RBF-kernel SVM and varies C .

Table 4: Comparison of several SVM methods on the same train-test split.

Method	Kernel	Test Acc.	Precision	Recall	F1
Linear SVM	linear	0.8667	0.9231	0.8000	0.8571
Polynomial SVM	poly	0.7667	1.0000	0.5333	0.6957
RBF SVM	rbf	0.9000	0.9286	0.8667	0.8966

3.5 Graphic Demonstration

Decision Boundaries of RBF-SVM under Different C Values

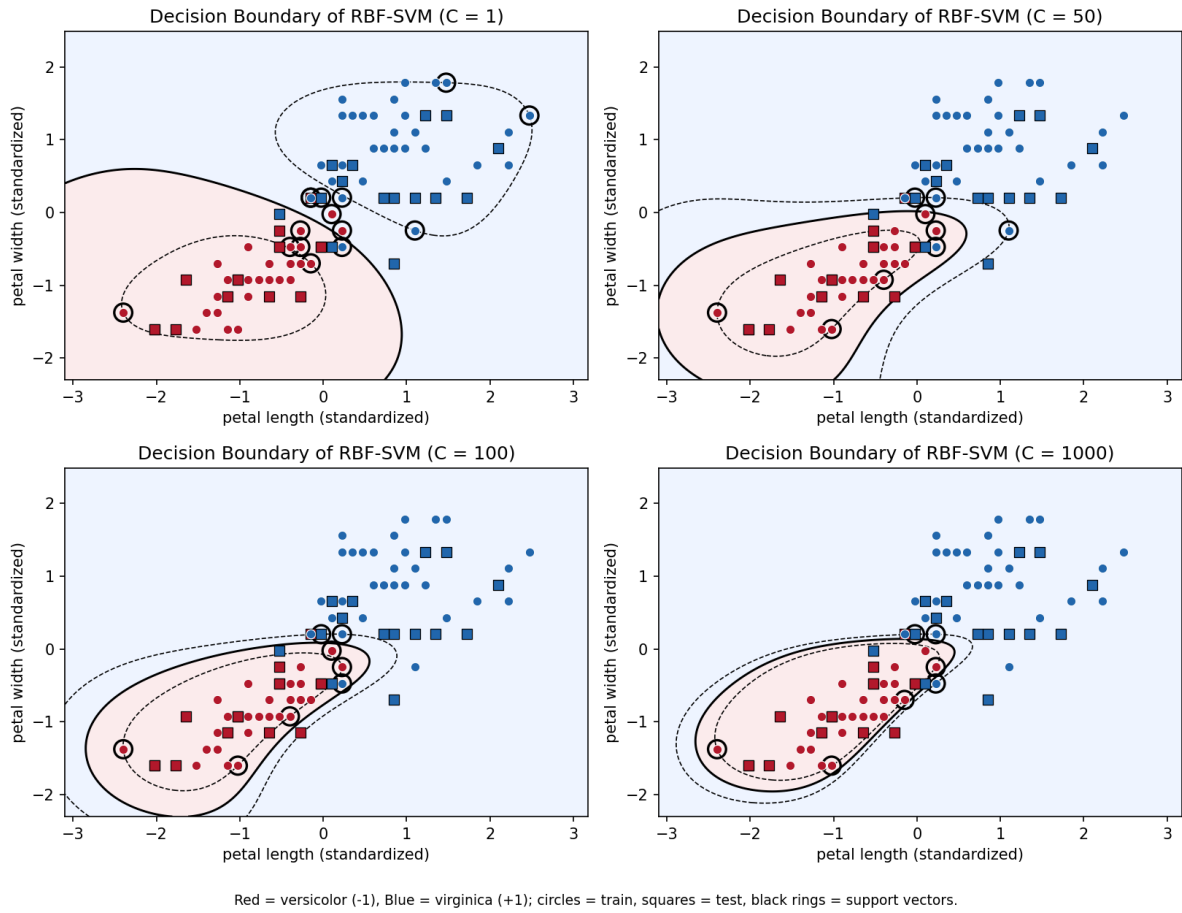


Figure 1: Decision boundaries and support vectors for $C = 1, 50, 100, 1000$. Red region / points: *versicolor* (-1); blue region / points: *virginica* (+1). Circles are training samples, squares are test samples, and black rings mark support vectors. The solid line is the decision boundary; dashed lines indicate the margins.

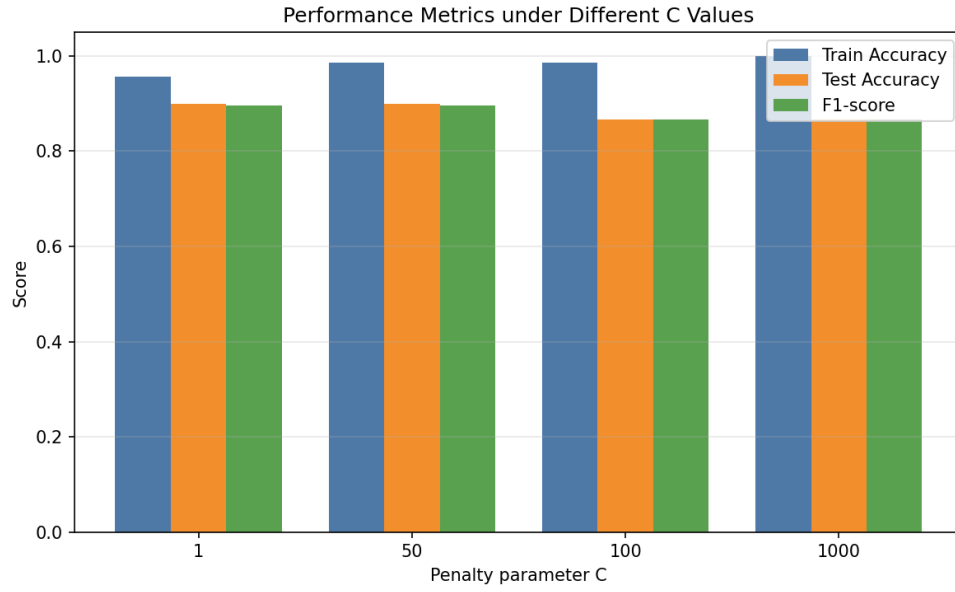


Figure 2: Train accuracy, test accuracy, and F1-score under different C values.

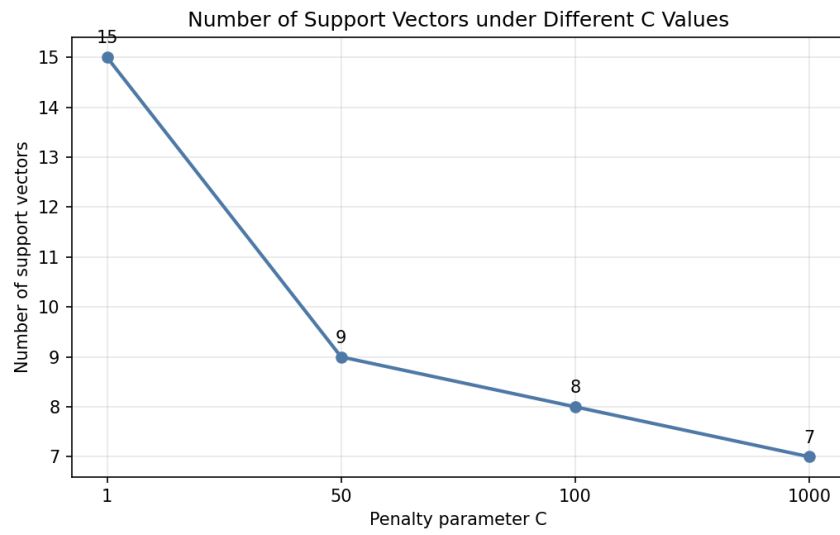


Figure 3: Number of support vectors under different C values.

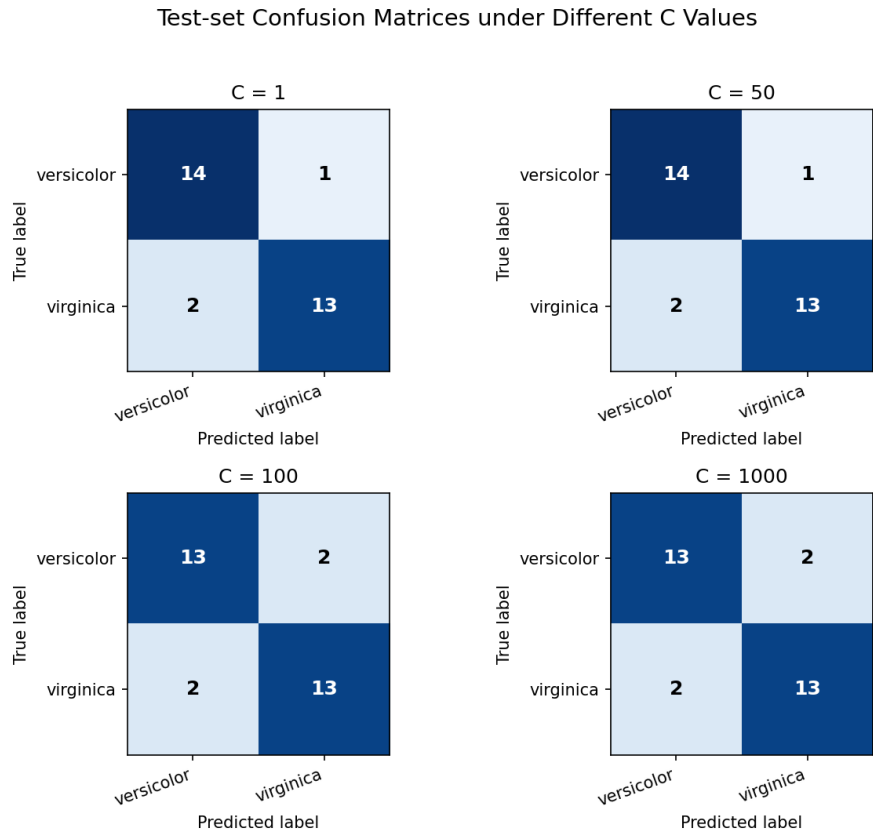


Figure 4: Test-set confusion matrices under different C values.

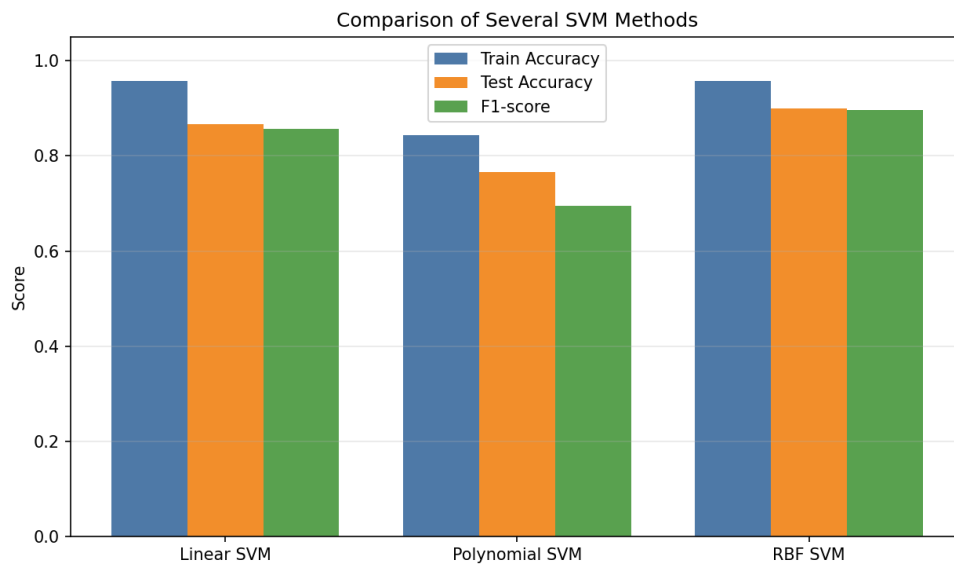


Figure 5: Comparison of linear, polynomial-kernel, and RBF-kernel SVM.

4 Discussion

4.1 Effect of C on Training Performance and Support Vectors

The main experiment controls all variables except C . The dataset, selected classes, features, train-test split, scaler, kernel, and gamma setting remain fixed. Differences in results can therefore be interpreted as the effect of changing the penalty parameter under this specific RBF-SVM setup.

As C increases, the model penalises margin violations more strongly. This is reflected in the training accuracy, which rises from 0.9571 at $C = 1$ to 1.0000 at $C = 1000$. Simultaneously, the number of support vectors decreases from 15 to 7. A larger C makes the classifier depend on fewer boundary-critical samples and fit the training set more tightly, which is consistent with the theoretical expectation that stronger regularisation leads to a narrower margin and fewer support vectors.

4.2 Effect of C on Test Performance and Generalisation

The test performance does not improve monotonically. $C = 1$ and $C = 50$ both achieve test accuracy 0.9000 and F1-score 0.8966. When C increases to 100 and 1000, test accuracy decreases to 0.8667 and F1-score decreases to 0.8667. This suggests that a stronger training penalty does not necessarily improve generalisation. In this run, very large C values make the classifier fit the training set more tightly but do not produce a test-set benefit.

The confusion matrices make this pattern more concrete. For $C = 1$ and $C = 50$, the model misclassifies 3 of the 30 test samples: one *versicolor* sample is predicted as *virginica*, and two *virginica* samples are predicted as *versicolor*. For $C = 100$ and $C = 1000$, the model misclassifies 4 test samples. This supports the metric table and shows that larger C values do not reduce the test-set errors in this split.

The decision-boundary figure (Figure 1) supports this interpretation. Smaller C gives a smoother and more tolerant boundary. Larger C adjusts more strongly to individual training samples. Because the dataset is small and the test set contains only 30 samples, small changes in the decision boundary can affect the reported test accuracy. The conclusion is therefore limited to this dataset, feature subset, train-test split, and fixed RBF-kernel setting.

4.3 Comparison of SVM Methods

The supplementary method comparison explains why RBF-SVM is used as the main model for the parameter experiment. Linear SVM performs reasonably well, but its test accuracy and F1-score are slightly lower than those of RBF-SVM. The polynomial-kernel SVM performs worse on this feature subset, especially in recall, which indicates that it misses more *virginica* samples. RBF-SVM achieves the best test accuracy and F1-score

among the three compared SVM methods, so it is a suitable focus for the detailed C analysis.

4.4 Limitations

This experiment uses a fixed single split rather than cross-validation, so the test metrics are sensitive to how the 30 test samples happen to be distributed. The Iris dataset is also small, and the two chosen classes are already reasonably well separated in the petal-length/petal-width space. These factors limit how strongly conclusions can be generalised beyond this specific setup.

5 Additional Analysis: $C \times \gamma$ Grid Search

This section is an optional extension and is separate from the main experiment. It examines whether changing γ together with C affects validation behaviour, using 5-fold cross-validation on the training set ($C \in \{1, 50, 100, 1000\}$, $\gamma \in \{0.01, 0.1, 1, 10\}$).

Table 5: $C \times \gamma$ grid search results (5-fold CV on training set).

C	γ	CV Mean	CV Std	Test Acc.	F1
1	0.01	0.9571	0.0350	0.8667	0.8571
1	0.1	0.9571	0.0350	0.9000	0.8966
1	1.0	0.9571	0.0350	0.9000	0.8966
1	10.0	0.9286	0.0639	0.9333	0.9333
50	0.01	0.9571	0.0350	0.8667	0.8571
50	0.1	0.9429	0.0535	0.8667	0.8571
50	1.0	0.9429	0.0535	0.8667	0.8667
50	10.0	0.9571	0.0571	0.9333	0.9333
100	0.01	0.9429	0.0535	0.8667	0.8571
100	0.1	0.9429	0.0535	0.8667	0.8571
100	1.0	0.9143	0.0833	0.8667	0.8667
100	10.0	0.9571	0.0571	0.9333	0.9333
1000	0.01	0.9429	0.0535	0.8667	0.8571
1000	0.1	0.9429	0.0535	0.9000	0.8966
1000	1.0	0.9143	0.0833	0.8667	0.8667
1000	10.0	0.9571	0.0571	0.9333	0.9333

The best cross-validation result is $C = 1$, $\gamma = 0.01$ (CV accuracy = 0.9571). This does not replace the main C -only comparison; it only shows that RBF-SVM performance depends on both C and γ , and that parameter selection should be based on cross-validation rather than training accuracy alone.

6 Conclusion

This experiment satisfies the three-step SVM experiment structure. Step 1 described the Iris dataset, the binary classification task (*versicolor* vs. *virginica*), the selected features, and the preprocessing pipeline. Step 2 explained several SVM methods, stated all fixed control variables, provided the mathematical formulations, and gave a pseudocode description of the procedure. Step 3 compared $C = 1, 50, 100, 1000$ on a fixed split and reported results using tables, confusion matrices, and figures.

The results show that increasing C improves training accuracy and reduces the number of support vectors, but does not guarantee better test performance. In this run, $C = 1$ and $C = 50$ achieve the best test accuracy among the four values. The penalty parameter should therefore be selected through validation and careful analysis rather than by assuming that the largest C is always best. The supplementary method comparison also shows that RBF-SVM is the strongest of the three compared SVM methods for this dataset and feature subset.